

Design and Evaluation of CFC-targeted Security Nudges

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ABSTRACT

Aimed at breaking up the myth of the “average user”, the notion of personalization promises to solve the contradiction between the population’s heterogeneity and one-size-fits-all security nudges. To further explore the promising avenue, we propose to design personalized security nudges targeted at different mindsets and make the Consideration for Future Consequences (CFC) the testbed. Namely, we designed two CFC-targeted security nudges, Promotion and Prevention, for the individuals who care about future and immediate consequences, respectively. An online survey (N = 145) was conducted to test their effectiveness. Results show that both the nudges can improve users’ security attitudes, while the moderation effects imply that the Promotion nudge is merely effective for the users having deep concerns about future consequences. The findings indicate the feasibility of designing security nudges targeted at future orientations and illustrate the importance of tailoring nudges according to individuals’ differences.

CCS CONCEPTS

• **Security and privacy** → **Usability in security and privacy**;
• **Human-centered computing** → **Empirical studies in HCI**;
Empirical studies in interaction design.

KEYWORDS

nudge in cybersecurity, personalization, consideration for future consequences, moderation effects

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1 INTRODUCTION

Nudge, a notion introduced by Thaler and Sunstein [23], implies leveraging some modifications of any aspect of the choice architecture to alter people’s decisions predictably. When used for social good and in an ethical way, nudges can help the public obtain optimal choices without any deprivation of freedom of choice. Furthermore, they generally cost less than economic incentives, while some elaborate nudges are more effective than great economic incentives reportedly [22]. An increasing number of usable security researchers are currently utilizing nudges to improve the effectiveness of security policies, and some remarkable achievements have been made [1]. By telling users concise privacy-relevant information and meaningful actions, Almuhammedi et al. [2] succeeded in motivating them to manage privacy settings more effectively. Similarly, social proof, which shows users that their friends have used some security measures, can also be leveraged to raise users’ security awareness and empower them to adopt the measures [5].

Nonetheless, people make decisions differently, which means that a certain nudge effective for one group of people might not work for another group [6]. For example, according to Costa and Kahn [4], when used to reduce electricity bills, the aforementioned “social-proof” nudge did not function in conservative households so much as in liberal ones. Similar things happen in cyberspace. It has been discovered that password meter, which is frequently used to improve password strength, is only effective for certain personality traits [18]. Thus, under the premise of populations’ heterogeneity, the current one-size-fits-all security nudge mechanism can only reach local maxima [6, 18]. Namely, given a certain security nudge, only a portion of people will be affected and benefit from it, whereas the rights of the rest are ignored. To solve the problem, Egelman and Peer [6] have introduced the notion of personalization into cybersecurity, which is to provide different security nudges to people with different psychological traits.

Lots of efforts have been put into personalized security nudge design in recent years. By exploring the relationship between security attitudes and some well-studied psychometrics, Egelman and Peer [6] proposed and stressed the urgent need of personalizing security mitigations to individual differences. Several years later, Peer et al. [18] succeeded in encouraging their participants to choose stronger passwords through personalized security nudges. The studies collectively illustrate the importance of personalized security nudges, which can be a promising avenue for future research. However, the research started comparatively later, and there is still

much work to be done. Most of the existing work [6, 11, 18] focuses on correlating frequently-used security nudges with some psychological traits and leveraging the correlations to allocate the nudges in a personalized manner. The schemes indeed work under some circumstances, but sometimes the results can not be directly and theoretically explained. A typical example is that the social-proof (or social-norm) nudge only affects the users who score medium in the NFC (Need for Cognition) scale [18].

Actually, individuals with several psychological traits usually have corresponding mindsets [3, 13]. In other words, people with certain psychological traits expect corresponding stimuli from the outside world [15]. For example, narcissistic people tend to look for others' compliments [3]. Thus, we propose to target nudge design at the mindsets that people have, providing the stimuli they expect, as an alternative scheme to improve the effectiveness of personalized security nudges further. Moreover, motivated by the relatively abundant research [6, 16, 18] on the correlations between the CFC (Consideration for Future Consequences) scores and security behaviors, we chose CFC as a starting point. We designed two personalized security nudges by providing the information that individuals with different CFC scores would like to see. The CFC scale we used was CFC-14 [13], which was composed of two sub-scales, CFC-Immediate and CFC-Future. Thus, one (called Prevention) of the two nudges was targeted at the individuals who had high CFC-Immediate scores, and the other (called Promotion) was for the ones who scored high on the CFC-Future sub-scale. To test the potential of the scheme in inducing essential behavioral changes, we decided to conduct an online survey with 145 participants to test its impact on security attitudes first, which can be regarded as a pre-test of the future field experiment. The survey results indicated that both the nudges were better in improving security attitudes than the control group. Leveraging the Johnson-Neyman technique, we discovered that participants' CFC-Future scores moderated the impact of the Promotion nudge, which implied the need to allocate the Promotion nudge to the individuals with high CFC-Future scores. There was no evidence, however, of the moderation effects of the CFC-Immediate scores on the effectiveness of the Prevention nudge.

2 RELATED WORK

Our research builds on prior work on security-related psychological traits and personalized security nudges.

2.1 Security-related Psychological Traits

A significant premise for designing and implementing personalized security nudges is to explore the psychological traits related to security attitudes and behaviors. The "Big Five" model is one of the personality models frequently used and studied by usable security researchers. The "Big Five" model identifies five factors in human personality traits: openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism [9]. In the online survey conducted by McCormac et al. [17], open and agreeable participants usually had high security awareness. Moreover, Gratian et al. [10] revealed that extraversion was also a predictor of good security behaviors. However, a series of experiments conducted by Egelman and Peer [7] suggested that the "Big Five" model was a weak

predictor of security attitudes. Thus, they tried other psychometrics to find the psychological traits that were strongly connected with security attitudes, such as DoSpERT (Domain-Specific Risk-Taking Attitudes), GDMS (General Decision-Making Styles), and CFC (Consideration for Future Consequences) [7]. As a result, they found statistically significant connections between security attitudes and all the three psychometrics mentioned above [7]. Apart from the psychological traits mentioned, many other traits have been found to have the predictive capability of security awareness, including sensation-seeking [12] and cognitive processing characteristics [14].

In summary, the abundant research on the connections between psychological traits and security behaviors has empowered us to design security nudges targeted at the traits.

2.2 Personalized Security Nudges

Since introduced in 2015 [6], the concept of "personalized security nudges" has received much attention from researchers and practitioners. Malkin et al. [15] customized HTTPS warnings for individuals' differences in decision-making, which were measured by the General Decision-Making Styles scale (GDMS). Although the connections between the effectiveness of a few personalized security warnings and certain decision-making styles were found, most of the nudges did not work as expected [15]. Personalized security nudges were also used to enhance password strength. Leveraging the moderation effects of certain psychological traits on the effectiveness of several common security nudges, Peer et al. [18] attempted to match the nudges and the populations who were most likely to be affected by them. The results suggested that the personalization scheme motivated users to create more secure passwords than a "one-size-fits-all" scheme. Similarly, Guo et al. [11] also designed a personalized scheme called DPPP (Dynamic Personalized Password Policies) to prevent users from choosing weak passwords, which did a better job than two standard password composition policies, Basic8 and 3class8. In the context of privacy policies, Schöning et al. discovered that when presentation format matched users' cognitive styles, their privacy concerns and risk perceptions were lowered, but disclosure behaviors were not affected [21].

The paragraph above is a relatively comprehensive summary of the existing work on personalized security nudges to the best of our knowledge. It can be concluded that the history of the research is not long, and there have not been many successful attempts that induced substantial behavioral changes. Thus, mysteries still exist, and much work needs to be done. For example, the measure adopted by Malkin et al. [15] that designed nudges targeted at the mindsets that some individuals' had is still a promising area, despite the insignificant results, and needs more in-depth explorations.

3 NUDGE DESIGN

Motivated by the innovative design idea but unsatisfactory results in the study conducted by Malkin et al. [15], we decided to try the idea again and explore further, providing the information (stimuli) that users with certain psychological traits generally look for. Moreover, the Consideration for Future Consequences (CFC) was chosen because of two reasons. On the one hand, many studies [7, 16, 18] have proved its strong relationship with security behaviors and

attitudes. On the other hand, little work has leveraged it to design personalized security nudges. Thus, in our study, we made CFC the testbed.

According to Joireman et al. [13], there are two distinguishing factors in individuals' CFC: one is about the concerns with future consequences (noted as CFC-Future), and the other is about immediate consequences (noted as CFC-Immediate). Furthermore, they discovered that individuals' CFC scores were linked with their regulatory focus orientation [13]. Namely, the people who have many concerns about future consequences (CFC-Future) generally adopt a "Promotion" orientation. The notion of "ideal self" appeals to them, and they take actions to obtain gains [13]. By contrast, the people who score high on CFC-Immediate prefer a "Prevention" orientation. They think "ought self" is motivating, and what nudges them to take actions is to avoid losses [13].

Following the link between CFC scores and the preference for different future orientations, we designed two different security nudges for individuals with high CFC-Future scores and CFC-Immediate scores, respectively. For the people with high CFC-Future scores, the advantages of adopting security measures were framed as "reflects high security awareness" and "be among the champions in cybersecurity". It is a typical example of the Promotion orientation. For the people who scored high on CFC-Immediate, we described the aim to adopt security measures as "avoid terrible security accidents", which can be regarded as a Prevention orientation.

Motivated by the practice in the research conducted by Ur et al. [24] that recommended an enhanced password, the security measure we tested and wanted a user to adopt was an alternative password created by modifying the user's previous password. Finally, the two CFC-targeted security nudges were designed as follows.

- **Promotion:** Setting a strong password is what security experts recommend to effectively reduce the security risk of identity theft, which reflects high security awareness. Choose the improved version of your previous password that we created for you, and you will be among the champions in cybersecurity.
- **Prevention:** Recent studies show that most identity theft incidents are caused by weak and easy-to-guess passwords. Choose the improved version of your previous password that we created for you, and you can largely avoid such terrible security incidents.

4 EVALUATION

A between-subjects survey was conducted with 145 participants, through which their CFC scores and security attitudes were obtained. Utilizing rounds of data analysis, we compared participants' security attitudes in each nudge group to the control group to evaluate the effectiveness of the nudges and measured the moderation effects of CFC scores.

4.1 Survey Design

Identified as a pretest of future research on substantial behavioral changes, the online survey only measured self-reported security attitudes by nature. It was composed of three parts: the CFC-14 scale

(introduced later), a security decision problem, and demographic questions. The participants needed to finish the CFC-14 scale at first, after which they were shown a security decision problem that asked the degrees of willingness to adopt the alternative password recommended by us. Finally, they needed to answer the demographic questions before they exited. The survey was in Chinese since it was conducted in China.

In our study, the CFC-14 [13] scale was chosen to measure individuals' considerations for future consequences. According to Joireman et al. [13], the scale consists of two sub-scales, which measure the two distinguishing factors in individuals' CFC, CFC-Future and CFC-Immediate, respectively. Both the sub-scales are composed of seven items, and all the items take the form of a seven-point Likert scale. Therefore, an individual's CFC-Future and CFC-Immediate scores are both obtained by calculating the average of the responses to the corresponding seven items. A higher score implies more concerns about the corresponding aspect of consequences.

The CFC-14 scale was initially written in English, and there were few Chinese versions that we could get. Thus, we translated the scale by ourselves, which took three rounds. At first, two bilinguals translated the scale independently, after which they had a discussion about inevitable disagreements and culture-specific issues and formed a draft. The draft was then handed to the team leader, who was also a bilingual and was responsible for revising it. In the final round, we applied the revised version to a small collection of people, and they provided several pieces of advice to help us fix some hard-to-understand sentences.

The part "security decision problem" was where security attitudes were measured and the CFC-targeted nudges were tested. There were three security decision problems in total, and each participant could only see one of the three randomly. In other words, the survey was a between-subjects study with a random assignment, and the experimental condition was the three different security decision problems. Two of the three were based on the "Promotion" and "Prevention" nudges mentioned above. Moreover, another one was the control group (noted as "Control"), in which simple prompts were provided. We tried to simulate a real security scenario by asking the participants to imagine setting passwords for an online banking account. The three security decision problems (translated into English) are shown in Table 1, which all showed up with the same illustrating picture in the survey¹. Moreover, each decision problem was a six-point Likert scale. The participants expressed their degrees of willingness to adopt an improved password by choosing a point between 1 and 6 (1 = totally unwilling to adopt, 6 = totally willing to adopt), which is exactly what we call "security attitudes" in this paper.

After completing the CFC-14 scale and the security decision problem, the participants needed to respond to the demographic questions, including gender, age, and education.

¹The picture was designed according to the practice adopted by Ur et al. [24] (provided in the supplemental material).

Table 1: Security Decision Problems

Group	Scenario Description	Nudge	Question
Control (N = 47)	Suppose that you have just set a password for an online banking account, and a warning pops up like the figure below. The warning says that your password is too weak, and it recommends an improved version based on your previous password. (with an illustrating picture)	N/A	How willing are you to adopt the improved password? Please indicate your willingness. (1 = totally unwilling, 6 = totally willing)
Promotion (N = 52)		The Promotion Nudge	
Prevention (N = 46)		The Prevention Nudge	

4.2 Recruitment, Compensation, and Demographics

Participants were recruited by snowball sampling via WeChat posts, and all were Chinese. To avoid potential bias, we did not tell them that the real purpose of the study was to test security attitudes in the beginning. They only knew that it was a survey on the consideration for future consequences (CFC). However, everyone was debriefed of the deception after the study was over. Moreover, each participant was compensated RMB ¥8.00, or roughly USD \$1.22.

A total of 150 participants were recruited, but five failed to pass the attention check question (an image recognition test). So, the sample consisted of 145 participants. Among the participants, (1) 70 self-identified as male, 74 as female, and one as non-binary; (2) most aged 18 - 29 (47.6%) and 30 - 39 (31.0%); (3) all had received secondary school education, and most (67.6%) held a bachelor degree. The participant distribution among the three experimental conditions is presented in Table 1.

4.3 Analysis

After the survey, we ran several rounds of analysis to test the effectiveness of the two CFC-targeted nudges and the moderation effects of CFC scores. At first, the reliability and validity of the two CFC sub-scales were checked via the Confirmatory Factor Analysis and the Cronbach's α . As a result, it was discovered that the two sub-scales had good reliability (Cronbach's $\alpha_F = .807$, Cronbach's $\alpha_I = .843$), acceptable convergent validity ($AVE_F = .370$, $CR_F = .798$; $AVE_I = .423$, $CR_I = .831$), and good discriminant validity ($\sqrt{AVE_F} = .608$, $\sqrt{AVE_I} = .650$, both were larger than the absolute value of the Pearson correlation coefficient between the two CFC factors).

The results of the Kolmogorov-Smirnov test indicated that none of the participants' security attitudes, CFC-Future scores, and CFC-Immediate scores conformed to a Gaussian distribution. Thus, we leveraged the Kruskal-Wallis test and the Mann-Whitney U test to compare the three dimensions among the three experimental conditions. Also, the Spearman correlation analysis was run to unravel the correlation between CFC scores and security attitudes. To explore the moderation effects of CFC scores, we chose the Johnson-Neyman technique [19]. With the technique, we could identify the range of CFC-Future or CFC-Immediate scores where

the CFC-targeted nudges significantly affected participants' security attitudes.

5 RESULTS & DISCUSSIONS

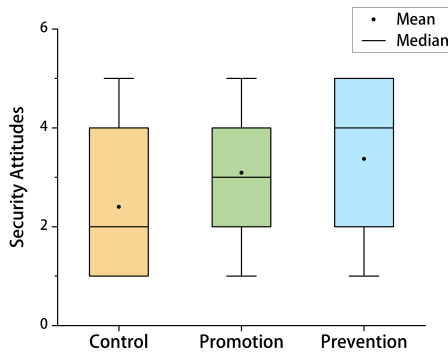
5.1 Results

The Kruskal-Wallis test revealed no significant differences in participants' CFC-Future and CFC-Immediate scores across all three experimental conditions. As Figure 1(a) suggests, the participants in the two nudge (Promotion and Prevention) groups were significantly more willing to adopt the alternative password ($p = .008$, by the Kruskal-Wallis test) than the control group. Although it seems that the Prevention nudge was better at improving security attitudes than the "Promotion" nudge according to Figure 1(a), the difference between the two was not significant ($p = .314$, by the Mann-Whitney U test).

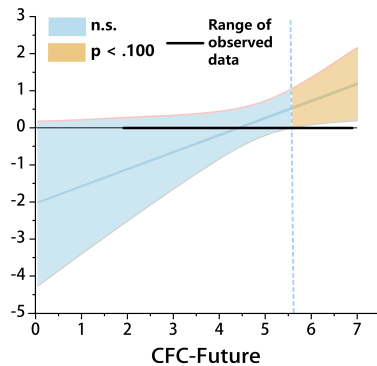
We ran the Spearman correlation analysis in the two nudge groups to find the correlations between CFC scores and security attitudes. In the Promotion group, CFC-Future scores were positively correlated with security attitudes ($\rho_s = .307$, $p = .023$). By contrast, no significant correlations were discovered between participants' CFC-Immediate scores and security attitudes in the Prevention group. Motivated by the significant relationship discovered in the Promotion group, we leveraged the Johnson-Neyman technique to explore the moderation effects of CFC-Future scores on the impact of the Promotion nudge. As a result, it was discovered that individuals' concerns about future consequences (CFC-Future scores) indeed had (marginally) significant moderation effects on the impact of the Promotion nudge on security attitudes ($p = .057$). According to Figure 1(b), at the significant level $\alpha = .100$, the Promotion nudge had a (marginally) significant impact on the participants who had CFC-Future scores higher than 5.535.

5.2 Discussions

As mentioned above, we designed two CFC-targeted nudges, Promotion and Prevention, based on the typical manifestation of the individuals who had high CFC-Future and CFC-Immediate scores, respectively. The two nudges were both more effective in motivating the participants to adopt the alternative password than the control group. It can be implied that both the Promotion and Prevention orientations contribute to the improvement of individuals' security attitudes. According to Joireman et al. [13], the individuals



(a) Distribution of Security Attitudes



(b) Moderation Effects of CFC-Future Scores on the Impact of the Promotion Nudge

Figure 1: Survey Results

who prefer Promotion orientation generally pursue ideals, hopes, and aspirations, and they take actions to obtain the harvest; the individuals who favor the Prevention orientation, by contrast, focus on oughts, duties, and responsibilities, and they take actions to avoid potential losses. Thus, the researchers and practitioners who want to promote security measures and improve users' security intentions can leverage the Promotion and Prevention orientations to design their ways of delivering cybersecurity-related information, such as security warnings and privacy notices, by emphasizing the "ideal self" users can be and the potential losses they can avoid.

However, the moderation effects of the CFC-Future scores revealed that the Promotion nudge was effective merely for the participants who deeply cared about the future consequences of their actions. As mentioned above, these people were eager to obtain gains and pursue a better "self", which was why they were motivated by the Promotion nudge. Nonetheless, the Promotion nudge was not effective for the participants who did not pay much attention to future consequences. The clear distinction further emphasizes

the importance of personalized security nudges, tailoring security nudges by individuals' different characteristics and needs.

The notion of the Prevention nudge and its success in improving security attitudes are not surprising because it can be regarded as a description of the terrible consequences that unsecure behaviors will result in, as suggested by previous studies [8, 20]. Despite its superiority over the control group in improving participants' security attitudes, we failed to find its relationship with participants' CFC-Immediate scores, i.e., the moderation effects of the CFC-Immediate scores on its impact. The results are partially supported by the research conducted by Joireman et al. [13], which indicates that individuals' concerns with immediate consequences are not related to their exercise and eating habits. Thus, based on the empirical results in the paper, we can not conclude about the personalization of the Prevention nudge according to individuals' CFC-Immediate scores.

In practice, the two CFC-targeted nudges proposed in the paper can be implemented independently by telling users what they can achieve and prevent if they adopt some security measures. Besides, as the methodology in the paper suggested, the two nudges can also act as the assistance of other nudges. For example, the data-driven password meter proposed by Ur et al. [24] will recommend a new password based on the previous password to users, and the two CFC-targeted nudges proposed by us can make the suggested improvement more appealing by emphasizing the potential gains and the losses that can be prevented.

5.3 Limitations

To make the protection that security mechanisms could provide specific and the adoption reasonable, we mentioned the security risks of identity theft concisely in the Promotion nudge instead of the vague expression "Security experts recommend setting a strong password". However, this seemed to make the Promotion nudge overlap with the Prevention one. Thus, in future iterations of the work, we plan to design an improved "Promotion" nudge and compare it with the current version to take a more in-depth look.

According to the p-value .057, the moderation effects of CFC-Future scores on the effectiveness of the Promotion nudge were marginally significant, which seemed unusual under the general significant level $\alpha = .050$. However, the survey was merely a pre-test with relatively small sample size, so we believe that the marginally significant results are acceptable.

Also, social media posts usually do not spread far, so our sample is not so general and representative as we expected.

6 CONCLUSION & FUTURE WORK

This paper explores the use of two CFC-targeted nudges, Promotion and Prevention, in improving users' security attitudes. The two nudges are designed according to the corresponding mindsets that users with deep concerns about future and immediate consequences generally have. To study their effectiveness in improving security attitudes, we conducted an online survey and compared them with the control group. Results indicate that showing the gains that can be obtained and the losses that can be avoided by adopting security measures can both nudge users towards better security attitudes.

However, the moderation effects of CFC-Future scores on the effectiveness of the Promotion nudge suggest that only individuals who are deeply concerned about the future consequences of their actions will be affected by the Promotion nudge. The discovery further emphasizes the importance of personalized security nudges based on individuals' psychological differences.

In the future, we plan to explore further the effectiveness of personalized security nudges, especially CFC-targeted ones, in affecting security behaviors by conducting field experiments. We will also test how to combine the CFC-targeted nudges with a few standard security practices, such as password meter.

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REFERENCES

- [1] Alessandro Acquisti, Idris Adjerid, Rebecca Balebako, Laura Brandimarte, Lorrie Faith Cranor, Saranga Komanduri, Pedro Giovanni Leon, Norman Sadeh, Florian Schaub, Manya Sleeper, et al. 2017. Nudges for privacy and security: Understanding and assisting users' choices online. *ACM Computing Surveys (CSUR)* 50, 3 (2017), 44. <https://doi.org/10.1145/3054926>
- [2] Hazim Almuhtimed, Florian Schaub, Norman Sadeh, Idris Adjerid, Alessandro Acquisti, Joshua Gluck, Lorrie Faith Cranor, and Yuvraj Agarwal. 2015. Your location has been shared 5,398 times!: A field study on mobile app privacy nudging. In *Proceedings of the 2015 CHI Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI 2015). ACM, New York, NY, USA, 787–796. <https://doi.org/10.1145/2702123.2702210>
- [3] Daniel R Ames, Paul Rose, and Cameron P Anderson. 2006. The NPI-16 as a short measure of narcissism. *Journal of Research in Personality* 40, 4 (2006), 440–450.
- [4] Dora L Costa and Matthew E Kahn. 2013. Energy conservation “nudges” and environmentalist ideology: Evidence from a randomized residential electricity field experiment. *Journal of the European Economic Association* 11, 3 (2013), 680–702.
- [5] Sauvik Das, Adam DI Kramer, Laura A Dabbish, and Jason I Hong. 2014. Increasing security sensitivity with social proof: A large-scale experimental confirmation. In *Proceedings of the 2014 ACM SIGSAC conference on computer and communications security*. ACM, Scottsdale, AZ, USA, 739–749.
- [6] Serge Egelman and Eyal Peer. 2015. The myth of the average user: Improving privacy and security systems through individualization. In *Proceedings of the 2015 New Security Paradigms Workshop* (Twente, Netherlands) (NSPW 2015). ACM, New York, NY, USA, 16–28. <https://doi.org/10.1145/2841113.2841115>
- [7] Serge Egelman and Eyal Peer. 2015. Predicting privacy and security attitudes. *ACM SIGCAS Computers and Society* 45, 1 (2015), 22–28.
- [8] Adrienne Porter Felt, Alex Ainslie, Robert W Reeder, Sunny Consolvo, Somas Thyagaraja, Alan Bettis, Helen Harris, and Jeff Grimes. 2015. Improving SSL warnings: Comprehension and adherence. In *Proceedings of the 2015 CHI Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI 2015). ACM, New York, NY, USA, 2893–2902. <https://doi.org/10.1145/2702123.2702442>
- [9] Samuel D Gosling, Peter J Rentfrow, and William B Swann Jr. 2003. A very brief measure of the Big-Five personality domains. *Journal of Research in Personality* 37, 6 (2003), 504–528.
- [10] Margaret Gratian, Sruthi Bandi, Michel Cukier, Josiah Dykstra, and Amy Ginther. 2018. Correlating human traits and cyber security behavior intentions. *Computers & Security* 73 (2018), 345–358.
- [11] Yimin Guo, Zhenfeng Zhang, Yajun Guo, and Xiaowei Guo. 2020. Nudging Personalized Password Policies by Understanding Users' Personality. *Computers & Security* 94, 3 (2020), 101801.
- [12] Juan Herrero, Alberto Uruñeña, Andrea Torres, and Antonio Hidalgo. 2017. My computer is infected: The role of users' sensation seeking and domain-specific risk perceptions and risk attitudes on computer harm. *Journal of Risk Research* 20, 11 (2017), 1466–1479.
- [13] Jeff Joireman, Monte J Shaffer, Daniel Balliet, and Alan Strathman. 2012. Promotion orientation explains why future-oriented people exercise and eat healthy: Evidence from the two-factor consideration of future consequences-14 scale. *Personality and Social Psychology Bulletin* 38, 10 (2012), 1272–1287.
- [14] Christina Katsini, Christos Fidas, Marios Belk, George Samaras, and Nikolaos Avouris. 2019. A Human-Cognitive Perspective of Users' Password Choices in Recognition-Based Graphical Authentication. *International Journal of Human-Computer Interaction* 35, 19 (2019), 1800–1812.
- [15] Nathan Malkin, Arunesh Mathur, Marian Harbach, and Serge Egelman. 2017. Personalized security messaging: Nudges for compliance with browser warnings. In *2nd European Workshop on Usable Security* (Paris, France) (EuroUSEC'17). Internet Society, Reston, VA, USA, 1–12. <https://doi.org/10.14722/eurosec.2017.23008>
- [16] Arunesh Mathur and Marshini Chetty. 2017. Impact of user characteristics on attitudes towards automatic mobile application updates. In *Thirteenth Symposium on Usable Privacy and Security* ({SOUPS} 2017). USENIX Association, Santa Clara, CA, USA, 175–193.
- [17] Agata McCormac, Tara Zwaans, Kathryn Parsons, Dragana Calic, Marcus Butavicius, and Malcolm Pattinson. 2017. Individual differences and information security awareness. *Computers in Human Behavior* 69 (2017), 151–156.
- [18] Eyal Peer, Serge Egelman, Marian Harbach, Nathan Malkin, Arunesh Mathur, and Alisa Frik. 2020. Nudge me right: Personalizing online security nudges to people's decision-making styles. *Computers in Human Behavior* 109, 12 (2020), 106347.
- [19] Kristopher J Preacher, Patrick J Curran, and Daniel J Bauer. 2006. Computational tools for probing interactions in multiple linear regression, multilevel modeling, and latent curve analysis. *Journal of educational and behavioral statistics* 31, 4 (2006), 437–448.
- [20] Leilei Qu, Cheng Wang, Ruojin Xiao, Wenchang Shi, and Liang Bin. 2019. Towards Better Security Decisions: Applying Prospect Theory to Cybersecurity. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI EA'19). ACM, New York, NY, USA, Paper No. LBW2613. <https://doi.org/10.1145/3290607.3312782>
- [21] Charlotte Schöning, Christian Matt, and Thomas Hess. 2019. Personalised nudging for more data disclosure? On the adaption of data usage policies format to cognitive styles. In *Proceedings of the 52nd Hawaii International Conference on System Sciences* (Maui, HI, USA) (HICSS '19). University of Hawaii at Manoa, Honolulu, HI, USA, 4395–4404. <https://doi.org/10.24251/HICSS.2019.532>
- [22] Cass R Sunstein. 2014. Nudging: a very short guide. *Journal of Consumer Policy* 37, 4 (2014), 583–588.
- [23] Richard H Thaler and Cass R Sunstein. 2009. *Nudge: Improving decisions about health, wealth, and happiness*. Penguin, New York, NY, USA.
- [24] Blase Ur, Felicia Alfieri, Maung Aung, Lujo Bauer, Nicolas Christin, Jessica Colnago, Lorrie Faith Cranor, Henry Dixon, Pardis Emami Naeini, Hana Habib, et al. 2017. Design and evaluation of a data-driven password meter. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, CO, USA) (CHI 2017). ACM, New York, NY, USA, 3775–3786. <https://doi.org/10.1145/3025453.3026050>